Profit

0
65
110
110
125

3. Backtracking

Backtracking is a general algorithmic technique that searches every possible combination in order to solve and optimization problem it is also known as depth first search (DFS) or branch and bound.

Backtracking is a general algorithm for finding all or some solutions for computational problem notatedly constraint specification problem that incrementally builds candidates to the solution.

Backtracking is the important tool for solving constraint specification problem such as cross word and seudoko problem.

N-Queen problem :

1. 4-Queen problem :

∴ We have 4 queen to place in 4×4 chess board. to place that 4 queens we have rules :

1. At same rows no repeat queen is allowed
2. At same column no repeat queen is allowed
3. At same diagonal no repeat queen is allowed.

∴ queens are 1, 2, 3, 4

	1	2	3	4	
1	1	×	×	×	at step-1 one queen is remaining so this solution is wrong. ∴ backtrack the problem
2	×	×	2	×	
3	×	×	×	×	
4	×	3	×	×	

	1	2	3	4
1	X	Q	X	X
2	X	X	X	Q
3	Q	X	X	X
4	X	X	Q	X

∴ In step-2 all queens are placed so this is right solution.

∴ Mirror image of solution is also a right solution.

	1	2	3	4
1	X	X	Q	X
2	Q	X	X	X
3	X	X	X	Q
4	X	Q	X	X

∴ We got two solutions:

Solution 1 =

Column →	1	2	3	4
	2	4	1	3
↓ Rows	1	2	3	4

Mirror Solution =

Columns →	1	2	3	4
	3	1	4	2
↓ Rows	1	2	3	4

2. 8-queen problem:

∴ We have 8 queens

queens are 1 2 3 4 5 6 7 8

∴ Some rules are applicable.

Step 1:

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	1	2	3	4	5	6	7	8
1	1	X	X	X	X	X	X	X
2	X	X	2	X	X	X	X	X
3	X	X	X	X	3	X	X	X
4	X	4	X	X	X	X	X	X
5	X	X	X	5	X	X	X	X
6	X	X	X	X	X	X	X	X
7	X	X	X	X	X	X	X	X
8	X	X	X	X	X	X	X	X

∴ In this step three queens are remaining So it is corong Solution.

	1	2	3	4	5	6	7	8
1	X	1	X	X	X	X	X	X
2	X	X	X	2	X	X	X	X
3	3	X	X	X	X	X	X	X
4	X	X	4	X	X	X	X	X
5	X	X	X	X	5	X	X	X
6	X	X	X	X	X	X	X	X
7	X	X	X	X	X	X	X	X
8	X	X	X	X	X	X	X	X

In this step still three queens are remaining So it is corong Solution

	1	2	3	4	5	6	7	8
1	X	X	1	X	X	X	X	X
2	2	X	X	X	X	X	X	X
3	X	X	X	3	X	X	X	X
4	X	4	X	X	X	X	X	X
5	X	X	X	X	5	X	X	X
6	X	X	X	X	X	X	X	X
7	X	X	X	X	X	X	X	X
8	X	X	X	X	X	X	X	X

Still it is corong Solution.

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	1	2	3	4	5	6	7	8
1	X	X	X	1	X	X	X	X
2	2	X	X	X	X	X	X	X
3	X	X	3	X	X	X	X	X
4	X	X	X	X	4	X	X	X
5	X	5	X	X	X	X	X	X
6	X	X	X	X	X	X	X	6
7	X	X	X	X	X	X	X	X
8	X	X	X	X	X	X	X	X

	1	2	3	4	5	6	7	8
1	X	X	X	X	1	X	X	X
2	2	X	X	X	X	X	X	X
3	X	X	X	3	X	X	X	X
4	X	X	X	X	X	4	X	X
5	X	X	5	X	X	X	X	X
6	X	X	X	X	X	X	X	X
7	X	6	X	X	X	X	X	X
8	X	X	X	X	X	X	X	7

two queens are remain

	1	2	3	4	5	6	7	8
1	X	X	X	X	X	1	X	X
2	2	X	X	X	X	X	X	X
3	X	X	3	X	X	X	X	X
4	X	X	X	X	4	X	X	X
5	X	X	X	X	X	X	5	X
6	X	6	X	X	X	X	X	X
7	X	X	X	X	X	X	X	X
8	X	X	X	X	X	X	X	X

one queen is remain

	1	2	3	4	5	6	7	8
1	X	X	X	X	X	X	1	X
2	2	X	X	X	X	X	X	X
3	X	X	3	X	X	X	X	X
4	X	X	X	X	4	X	X	X
5	X	5	X	X	X	X	X	X
6	X	X	X	6	X	X	X	X
7	X	X	X	X	X	X	X	X
8	X	X	X	X	X	X	X	X

So

int
nu
Find
Sur

Exa

1. W

Am

∴

∴ We

0 =

1 =

$x_3 = 0$

$x_4 = 0$

∴ We g

∴ Actu

Sum of Subset :

Suppose we have given n distinct positive integer that is number and sum of these numbers are also given then we have to find out subset of these numbers whose sum is given sum number.

Example :

1. $W = \{4, 3, 5, 2\}$ $m = 7$

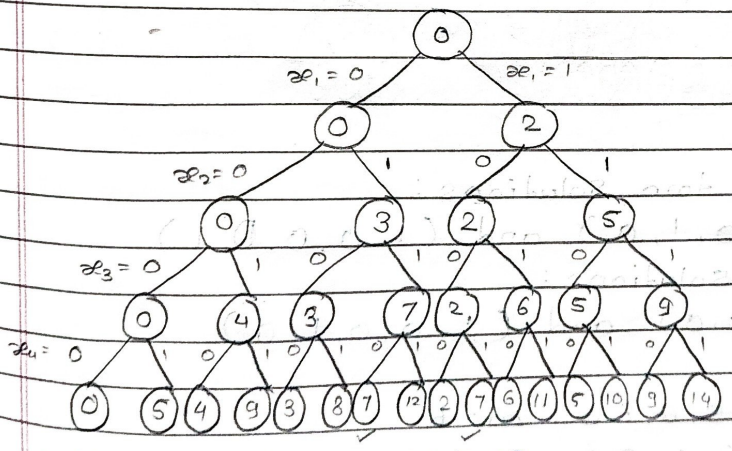
Arrange the given set in increasing order.

$\therefore W = \{2, 3, 4, 5\}$

We have to construct tree.

0 = Not Consider

1 = Consider $\therefore W = \{2, 3, 4, 5\} = \{x_1, x_2, x_3, x_4\}$



$\therefore$ We got two values.

(0, 1, 1, 0) and (1, 0, 0, 1)

$\therefore$ Actual solution :

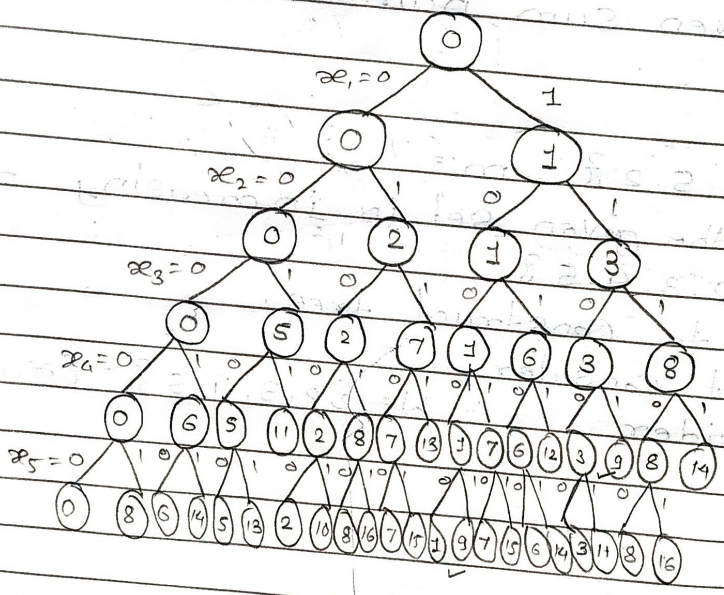
(1, 1, 0, 0) and (0, 0, 1, 1)

2. $S = \{1, 2, 6, 8, 5\}$ $m = 9$

Arrange set in increasing order.

$S = \{1, 2, 5, 6, 8\}$

$S = \{x_1, x_2, x_3, x_4, x_5\}$



∴ We got two solutions:

(1, 1, 0, 1, 0) and (1, 0, 0, 0, 1)

∴ Actual solutions:

(1, 1, 1, 0, 0) and (1, 0, 0, 1, 0)

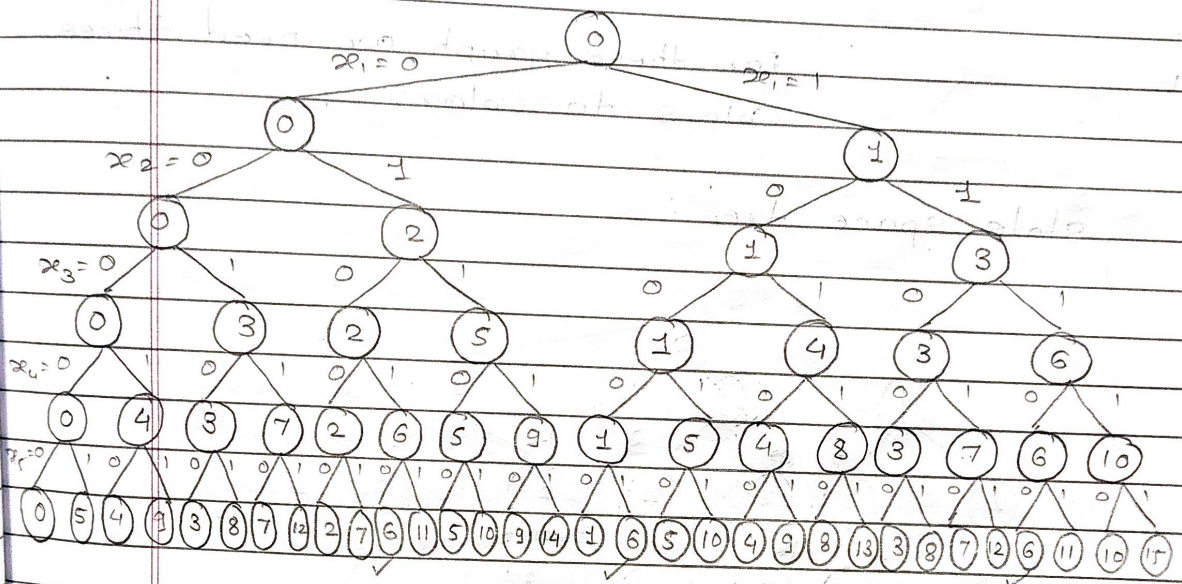
~~$S = \{3, 4, 5, 6, 8\}$ $m = 6$~~

3. $S = \{1, 2, 3, 4, 5\}$ $m = 6$

Given set is already in increasing order.

$S = \{1, 2, 3, 4, 5\}$

$S = \{x_1, x_2, x_3, x_4, x_5\}$



We got three Solutions.

(0,1,0,1,0) (1,0,0,0,1) (1,1,1,0,0)

Actual solutions are.,

(0,1,0,1,0) (1,0,0,0,1) (1,1,1,0,0)

Graph Coloring problem :

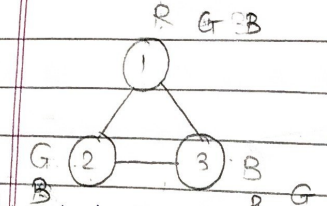
In the graph coloring problem we have given a graph and we have to color that graph for coloring that graph we have two conditions.

1. No ~~&~~ Same color is allowed to ~~the~~ its adjacent node.
2. To color the graph minimum no. of color is to be used.

Example.,

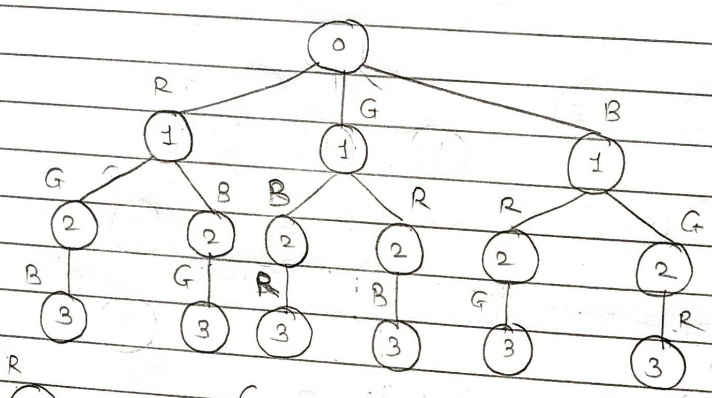
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1.

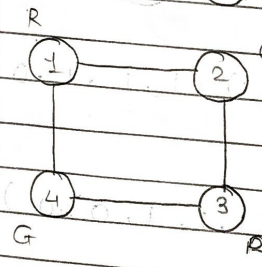


for these graph we need three Colors to Color.

State Space tree:

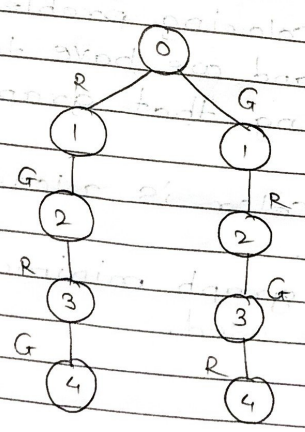


2.



For these graph we need two Colors to color.

State Space tree:



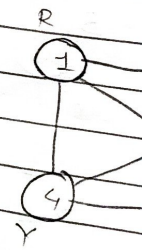
3.

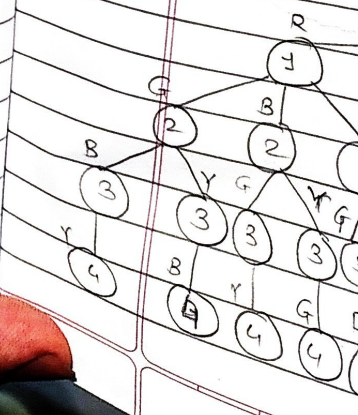


B G
R

Sta

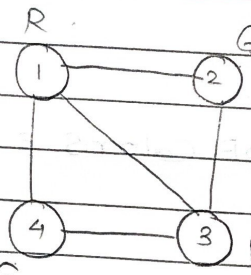
4.





B R
G B

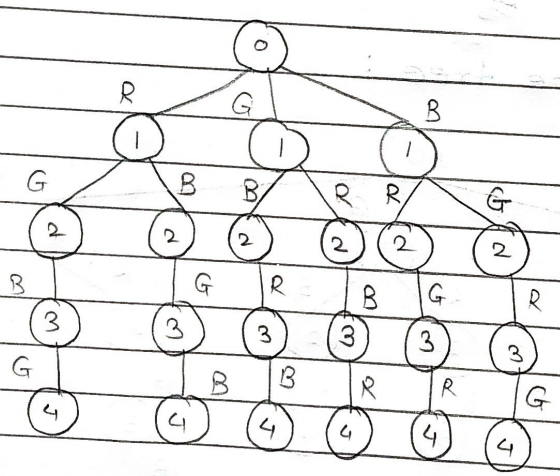
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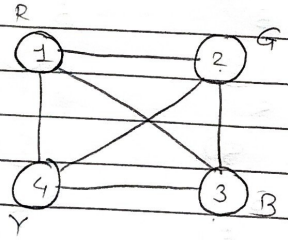
3. 

For these graph we need ~~three~~ four Colors to Color.

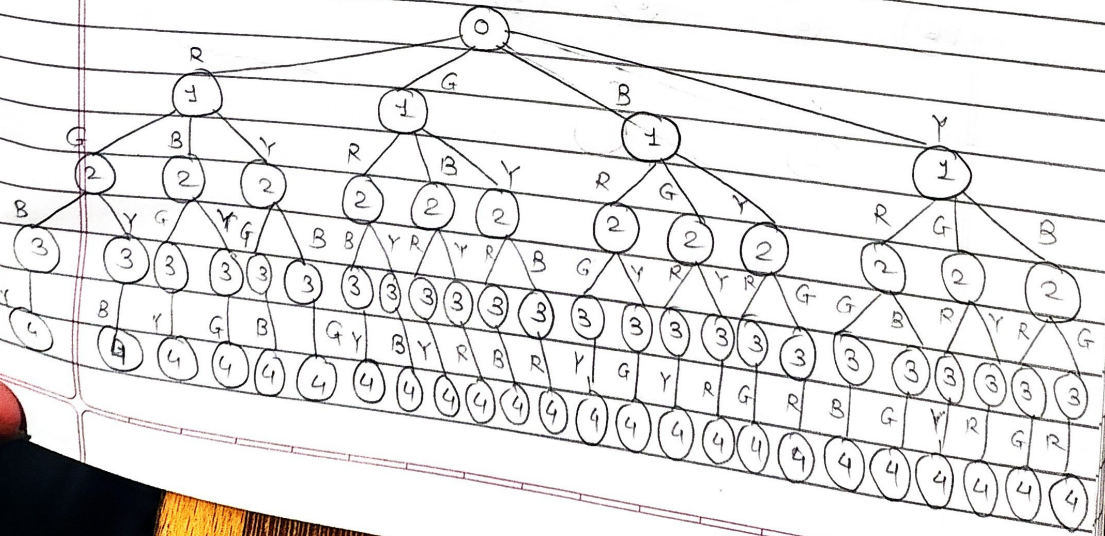
B G R G
R

State Space tree:



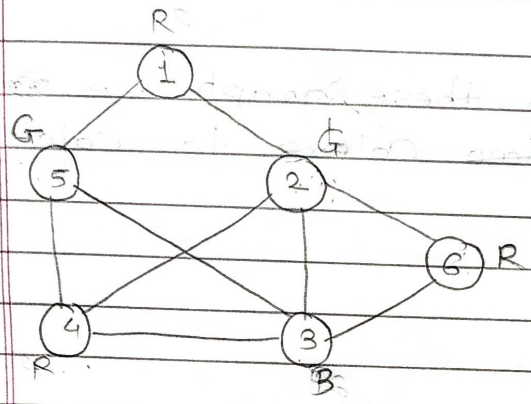
4. 

No. of Colours = 4



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5.



No. of Colors = 3

State Space tree :

